Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	
Project Name	Smart Lender:AI/ML-Based Loan Approval Prediction System

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard Python indentation (4 spaces) is followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into templates, static, model, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as `loan_amount`, `credit_history`, and `applicant_income` are self-explanatory.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as `predict()`, `train_model()`, and `load_model()` clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Important sections are commented, but additional comments can improve maintainability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Model training, prediction, and Flask routes are organized into separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code is minimized using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation is present; additional exception handling can

				further improve reliability.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas, NumPy)	Model training and prediction	High
2	Model Training Module	Python (Scikit-learn, XGBoost)	Training multiple ML algorithms	High
3	Saved XGBoost Model (`model.pkl`)	Pickle / XGBoost	Flask prediction module	High
4	Flask Prediction Route	Python (Flask)	Handles user input and prediction requests	High
5	HTML Form Templates	HTML, CSS, Bootstrap	User interface for loan prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with clear folder hierarchy.
Readability	4	Meaningful naming conventions and consistent formatting improve readability.
Reusability	5	Modular architecture allows components to be reused in future enhancements.
Documentation / Comments	4	Adequate documentation; more detailed comments could improve maintainability.
Overall Score	4.5/5	The project follows good coding practices with a modular design, readable code, and reusable components suitable for future development and deployment.